

TECHNICAL DOCUMENTATION FOR TEMPERATURE & WIND

SECTION 1 — SYSTEM CONTEXT & OBJECTIVES (TEMPERATURE & WIND MODULE)

1.1 Section Overview

This section documents the **Temperature and Wind Analytics Module**, which forms a core component of the overall Climate Data Analytics and BI system.

The purpose of this module is to process, transform, and analyze **historical meteorological data** at the station level, with a specific focus on:

- Air temperature behavior
- Wind characteristics (gust speed and direction)
- Long-term climate variability
- Extreme weather patterns (temperature-driven)

This module operates on **daily climate observations** and produces multiple analytical datasets used downstream in Business Intelligence reporting and predictive modeling.

1.2 Functional Scope of This Module

This module is responsible for the following system components:

✓ Data Processing Scope

- Cleaning and standardizing **daily temperature and wind observations**
- Handling missing and inconsistent meteorological data
- Converting raw station data into structured analytical datasets

✓ Analytical Scope

- Computing **temperature anomalies** using climatological baselines
- Detecting **heatwave events based on anomaly thresholds**
- Generating **monthly aggregated temperature and wind features**
- Supporting **trend analysis (°C per decade)**

✓ Forecasting Scope

- Providing input for **time-series forecasting models (SARIMA)**
- Preparing cleaned and structured temperature data for predictive modeling

✓ BI Integration Scope

- Delivering optimized datasets for:
 - Temporal trend visualization
 - Station-level comparison
 - Wind pattern analysis
 - Extreme event reporting

1.3 Temporal Scope

The analysis within this module covers:

Historical Analysis Period

1995 – 2025

This range is used for:

- Baseline computation
- Trend analysis
- Event detection

Forecasting Period

2026 – 2030

Used for:

- Temperature prediction
- Model validation and extension

1.4 Spatial Scope

The module focuses on **station-level meteorological data** within:

- Nova Scotia
- New Brunswick
- Newfoundland and Labrador
- Prince Edward Island

All observations are processed at the **individual station level**, enabling:

- High-resolution spatial analysis
- Regional comparison across stations

- Aggregation to higher geographic levels

1.5 Data Domain (Restricted Scope)

This section specifically uses the following variables:

Temperature Variables

Variable	Description
Max Temp (°C)	Maximum daily air temperature
Min Temp (°C)	Minimum daily air temperature
Mean Temp (°C)	Average of maximum and minimum temperature
Heat Degree Days (°C)	Heating demand indicator
Cool Degree Days (°C)	Cooling demand indicator

Wind Variables

Variable	Description
Spd of Max Gust (km/h)	Maximum recorded wind gust speed
Dir of Max Gust (10s deg)	Direction of strongest wind gust

1.6 Analytical Objectives of This Module

This module is designed to address the following analytical questions:

✓ Temperature Analysis

- How has **average temperature changed over time**?
- Are there **consistent warming trends across stations**?

✓ Anomaly Detection

- How do daily temperatures deviate from long-term normals?
- Are anomalies increasing over time?

✓ Heatwave Detection

- Are **extreme temperature events becoming more frequent**?
- How do heatwave duration and intensity vary spatially?

✔ Predictive Analysis

- What are the expected future temperature trends?
- How reliable are predictions based on historical patterns?

1.7 Outputs of This Module

This module produces the following datasets:

- Cleaned **daily temperature and wind tables**
- Monthly aggregated temperature and wind datasets
- Baseline climatology table
- Temperature anomaly dataset
- Heatwave event dataset
- Trend analysis dataset
- Forecasted temperature dataset

These outputs are consumed by:

- BI dashboards
- Analytical modeling layers
- Reporting pipelines

Perfect — now we move into the **most critical technical section** of your documentation. This must be **very precise, dataset-aware, and aligned with your notebooks + ECCC definitions**.

SECTION 2 — DATASET SPECIFICATION & DATA DICTIONARY (TEMPERATURE + WIND MODULE)

2.1 Section Overview

This section provides a **detailed technical specification of the dataset used in the Temperature and Wind Analytics Module**.

It includes:

- Column-level definitions
- Units and measurement standards
- Data types (raw and transformed)
- Dataset structure and constraints
- Observed data characteristics from preprocessing

The dataset is derived from **Environment and Climate Change Canada (ECCC) Daily Climate Observations**, which provide **station-level daily meteorological measurements** suitable for climate analysis and BI applications.

2.2 Dataset Structure

Granularity

Each record represents:

1 station $\times$ 1 day

Schema Composition

The dataset consists of:

- **Spatial attributes**
 - Station location and identifiers
- **Temporal attributes**
 - Date, Year, Month, Day
- **Meteorological variables**
 - Temperature and wind measurements

File Structure

The dataset follows a standardized naming convention:

climate_daily_{PROVINCE}_{CLIMATE_ID}_{YEAR}_P1D.csv

Where:

- PROVINCE = Province code (e.g., NS, NB)
- CLIMATE_ID = Unique station identifier
- YEAR = Observation year
- P1D = Daily resolution

2.3 Core Columns (Technical Data Dictionary)

2.3.1 Station & Metadata Fields

Column Name	Description	Unit	Raw Type	Final Type
Climate ID	Unique station identifier	N/A	string	category/string
Station Name	Official station name	N/A	string	string

Column Name	Description	Unit	Raw Type	Final Type
Latitude (y)	Station latitude	decimal degrees	float64	float64
Longitude (x)	Station longitude	decimal degrees	float64	float64

2.3.2 Temporal Fields

Column Name	Description	Unit	Raw Type	Final Type
Date / Date-Time	Observation date	YYYY-MM-DD	string	datetime64
Year	Year extracted from date	N/A	int	int32
Month	Month extracted from date	N/A	int	int32
Day	Day extracted from date	N/A	int	int32

2.3.3 Temperature Variables

Column Name	Description	Unit	Raw Type	Final Type
Max Temp (°C)	Maximum temperature observed during the day	°C	float64	float32
Min Temp (°C)	Minimum temperature observed during the day	°C	float64	float32
Mean Temp (°C)	Average of max and min temperature	°C	float64	float32
Heat Deg Days (°C)	Heating demand relative to 18°C baseline	°C	float64	float32
Cool Deg Days (°C)	Cooling demand relative to 18°C baseline	°C	float64	float32

2.3.4 Wind Variables

Column Name	Description	Unit	Raw Type	Final Type
Dir of Max Gust (10s deg)	Direction of strongest wind gust	Degrees ×10	float64	float32
Spd of Max Gust (km/h)	Speed of strongest wind gust	km/h	float64	float32

2.4 Measurement Definitions (Important Context)

✓ Mean Temperature

Mean Temp = (Max Temp + Min Temp) / 2

However, in real datasets:

- Sometimes missing → reconstructed during preprocessing

✓ Degree Days

- **Heat Degree Days (HDD):**
 - Measures how much heating is required below 18°C
- **Cool Degree Days (CDD):**
 - Measures cooling demand above 18°C

These are critical indicators for:

- Climate variability
- Energy demand patterns

✓ Wind Gust Measurements

- Only recorded at stations with wind sensors
- Missing values are **expected behavior**, not errors

2.5 Data Quality Characteristics (Observed in Your Dataset)

● Missing Values Pattern

From actual preprocessing:

Temperature:

- Fully null rows exist (~130K rows)
- Caused by:
 - Missing station recordings
 - Sensor downtime

Wind:

- High missing rate in direction field
- Cause:
 - Gust events not recorded daily
 - Direction only logged when gust occurs

✓ Interpretation Rule

NULL in wind $\neq$ missing data error

NULL in wind = no gust recorded

● Partial Null Rows

Observed pattern:

Max Temp = NULL

Min Temp $\neq$ NULL

Mean Temp = NULL

Handled using reconstruction logic (Section 3)

● Outliers

Example from your dataset:

Max Temp = 45°C (incorrect)

Cause:

- Sensor or recording error

Correction:

- Replaced with monthly mean value

2.6 Data Type Transformation (Detailed)

✓ Before Transformation

Type	Usage
float64	All numeric measurements
object/string	IDs and dates
string date	Raw date format

✓ After Transformation

Type	Purpose
float32	Reduces memory footprint (~50%)
datetime64	Enables time intelligence
category	Optimizes joins and filtering

✓ Memory Optimization Impact

- Dataset reduced significantly in size
- Improved:
 - Power BI performance
 - Query execution time
 - Model loading speed

2.7 Data Constraints

✓ Uniqueness Constraint

Primary Key = (Climate ID, Date)

Validated:

- No duplicate station-date records

✓ Baseline Constraint

For anomaly modeling:

Each (Station + Month + Day)

→ must have exactly one baseline value

Verified in dataset:

- No conflicting baselines

✓ Logical Constraints

- Mean Temp always exists (after cleaning)
- HDD and CDD cannot both be positive simultaneously
- Wind speed ≥ 0

2.8 Data Output for Downstream Processing

This dataset is transformed into:

- Daily temperature fact table
- Wind fact table
- Monthly aggregated tables
- Baseline climatology table

These outputs feed:

- Anomaly model
- Heatwave detection
- Forecasting model
- Power BI dashboards

✓✓ Why this section is strong

This is now:

- ✓ Based on your dataset + ECCC documentation
- ✓ Includes real **data issues you handled**
- ✓ Includes **real preprocessing logic context**
- ✓ Not generic definitions — actual system behavior
- ✓ Connects raw data → analytical usage

Excellent — this is now the **most important technical section** of your entire report. This section must reflect your **actual notebook logic + data engineering decisions**, not generic steps.

SECTION 3 — DATA CLEANING & TRANSFORMATION SPECIFICATIONS

3.1 Section Overview

This section documents the **complete data preprocessing and transformation pipeline** applied to the raw ECCC daily climate dataset for the Temperature and Wind Analytics module.

The transformations were implemented using **Python (Pandas-based workflows)** and were designed to:

- Ensure **data quality and consistency**
- Handle **missing and incomplete observations**
- Correct **invalid or anomalous values**
- Optimize data types for **BI performance**
- Prepare the dataset for **analytical modeling and reporting**

🔧 3.2 Raw Data Issues Identified

Before transformation, the dataset exhibited several **data quality challenges**, observed during exploratory analysis.

3.2.1 Fully Missing Temperature Records

Pattern observed:

Max Temp = NULL

Min Temp = NULL

Mean Temp = NULL

Heat Deg Days = NULL

Cool Deg Days = NULL

Cause:

- Missing sensor recordings
- Station inactivity
- Data ingestion gaps

Impact:

- These rows contain **no usable temperature information**

Resolution Rule

IF all temperature fields are NULL

→ DELETE row

Result:

- Approximately **130,000+ rows removed**
- Remaining dataset contains only **valid or partially usable records**

3.2.2 Partially Missing Temperature Data

Observed patterns:

Case 1:

Max Temp = NULL

Min Temp ≠ NULL

Mean Temp = NULL

Case 2:

Max Temp $\neq$ NULL
Min Temp = NULL
Mean Temp = NULL

✓ Imputation Strategy (Mean Temperature Reconstruction)

The Mean Temperature was reconstructed using the following logic:

IF Mean Temp IS NULL:

IF Max AND Min exist:

Mean Temp = (Max + Min) / 2

ELSE IF only Max exists:

Mean Temp = Max

ELSE IF only Min exists:

Mean Temp = Min

ELSE:

Remains NULL

Outcome:

- All valid rows now have **Mean Temperature populated**
- Ensures consistency for:
 - anomaly calculation
 - trend modeling

3.2.3 Invalid Data Patterns (Quality Validation)

A validation rule was applied to detect inconsistent records:

Invalid Pattern Checked:

Max Temp = NULL
Min Temp = NULL
Mean Temp = NULL
BUT

Heat Deg Days or Cool Deg Days $\neq$ NULL

Result:

- No such rows existed after cleaning
- Confirms internal dataset consistency

3.2.4 Outlier Detection and Correction

Example identified during analysis:

Station ID: 8204453

Date: 2008-04-20

Max Temp = 45°C

This value was significantly higher than typical observed values for that station.

✓ Correction Strategy

Replace outlier with:

Monthly average Max Temp for that station

Reason:

- Prevent distortion in:
 - anomaly calculation
 - trend estimation
 - forecast modeling

3.2.5 Wind Data Characteristics

Observed Issue:

- Frequent missing values in:
 - Gust direction
 - Gust speed

✓ Interpretation Rule

NULL wind values $\neq$ data error

NULL wind values = no gust recorded

✓ Cleaning Rule Applied

IF (Direction = NULL AND Speed = NULL):
REMOVE row

ELSE:

RETAIN row

Outcome:

- Retains meaningful wind observations
- Removes completely empty wind records

3.3 Data Transformation Steps

3.3.1 Date Standardization

Raw:

Date stored as STRING

Transformation:

Convert → datetime64 format

Purpose:

- Enable time-based operations:
 - filtering
 - grouping
 - time series modeling

3.3.2 Feature Extraction (Temporal Fields)

Derived fields:

- Year
- Month
- Day

Year = Date.year

Month = Date.month

Day = Date.day

3.3.3 Station ID Normalization

Raw:

Mixed-type (numeric + string)

Transformation:

Convert to STRING / CATEGORY

Purpose:

- Ensure consistent joins between tables
- Improve memory efficiency in BI model

3.3.4 Data Type Optimization (Critical Step)

Before:

Data Type	Issue
float64	High memory consumption
object	Slower joins
string date	Not usable for time modeling

After:

Column Type	Converted To	Benefit
float64	float32	~50% memory reduction
object	category/string	Faster filtering and joins
string date	datetime64	Time intelligence

✅ Impact

- Reduced dataset size significantly
- Improved Power BI performance
- Faster query execution

🔄 3.4 Aggregation Logic (Daily → Monthly)

3.4.1 Temperature Aggregation

Grouping keys:

Climate ID + Year + Month

Metrics calculated:

Monthly_Average_Max_Temp = mean(Max Temp)
Monthly_Average_Min_Temp = mean(Min Temp)
Monthly_Average_Mean_Temp = mean(Mean Temp)

Monthly_Max_Temp = max(Max Temp)
Monthly_Min_Temp = min(Min Temp)

3.4.2 Wind Aggregation

Aggregation rules (domain-specific):

Monthly_Max_Gust = MAX(Spd of Max Gust)
Monthly_Mean_Direction = MEAN(Dir of Max Gust)

Reason:

- Gust speed → extreme value metric
- Direction → average pattern indicator



3.5 Data Validation Checks



Uniqueness Check

Check duplicates:
(station_id, Date)

Result:

No duplicates found



Baseline Consistency Check

For:

station_id + month + day

Rule:

Only ONE baseline value

Result:

Validated — no conflicts

✓ Logical Constraints

- Mean Temp must exist after cleaning
- Heat Deg Days and Cool Deg Days cannot both be positive
- Wind speed ≥ 0

✓ 3.6 Final Output of Transformation Layer

After cleaning and transformation, the dataset is structured into:

✓ Daily-level datasets

- Clean temperature table
- Clean wind table

✓ Monthly-level datasets

- Monthly temperature aggregates
- Monthly wind aggregates

✓ Derived analytical datasets

- Baseline climatology
- Temperature anomaly table
- Heatwave event table
- Trend tables

Excellent — this is now the **core scientific + analytical heart of your system**. This section must clearly show that your project is **not just BI**, but also **climate modeling and time-series analysis**.

We'll structure it like a **technical modeling specification**, not a description.

SECTION 4 — BASELINE, ANOMALY & HEATWAVE MODEL SPECIFICATIONS

4.1 Section Overview

This section defines the **core analytical models** used in the Temperature and Wind Analytics module, including:

- Climatological baseline computation
- Temperature anomaly modeling

- Heatwave detection logic

These models transform cleaned observational data into **climate intelligence features**, which are used for:

- Trend analysis
- Extreme weather detection
- Forecast model input
- BI reporting

4.2 Baseline Temperature Model (Climatology)

4.2.1 Purpose

The baseline model establishes a **reference temperature for each station and day of the year**, allowing the system to measure deviations from "normal" climate conditions.

4.2.2 Definition

The baseline temperature is defined as:

$\text{Baseline}(s, m, d) = \text{mean}(T_mean(s, m, d, y)) \text{ over years } y \in [1995, 2025]$

Where:

- s = station
- m = month
- d = day
- y = year
- T_mean = observed mean temperature

4.2.3 Computation Logic

The baseline is calculated by grouping the cleaned dataset:

GROUP BY:

station_id, month, day

AGGREGATE:

mean(Mean Temp)

4.2.4 Output Structure

Resulting baseline table:

Column	Description
station_id	Station identifier
Month	Month (1–12)
Day	Day (1–31)
Baseline Mean Temp (°C)	Climatological average

✓ Key Properties

- Station-specific ✓
- Seasonal-cycle-aware ✓
- Consistent reference for anomaly modeling ✓

📄 4.3 Temperature Anomaly Model

4.3.1 Purpose

Temperature anomalies measure how much observed temperatures deviate from historical norms, enabling:

- Detection of warming trends
- Comparison across stations and seasons
- Identification of extreme conditions

4.3.2 Mathematical Definition

$$A(t, s) = T_{\text{obs}}(t, s) - T_{\text{baseline}}(s, m(t), d(t))$$

Where:

- $A(t, s)$ = temperature anomaly
- T_{obs} = observed mean temperature
- T_{baseline} = baseline temperature
- $m(t), d(t)$ = month and day derived from date

4.3.3 Computation Logic

Steps:

1. Join observed dataset with baseline table:

JOIN ON:
station_id
month
day

2. Compute anomaly:

$$\text{Temp_Anomaly} = \text{Mean_Temp} - \text{Baseline_Mean_Temp}$$

4.3.4 Output Table: fact_Daily_Anomaly

Column	Description
station_id	Station identifier
Date	Observation date
Baseline Mean Temp	Reference temperature
Temp Anomaly	Computed anomaly
TAnomaly_Key	Unique key

✅ Interpretation

Anomaly Value	Meaning
$A > 0$	Warmer than normal
$A < 0$	Cooler than normal
$A \approx 0$	Normal condition

✅ Statistical Characteristics (Observed)

From your dataset:

- Mean anomaly ≈ 0
- Distribution roughly symmetric
- Long tails represent extremes

🔥 4.4 Heatwave Detection Model

4.4.1 Purpose

The heatwave model identifies **persistent high-temperature events**, which are critical indicators of:

- Climate change
- Extreme weather risk
- Environmental stress

4.4.2 Formal Definition

A heatwave is defined as:

$A(t) \geq 2^{\circ}\text{C}$

FOR ≥ 3 consecutive days

4.4.3 Step-by-Step Detection Logic

✅ Step 1 — Flag Heatwave Candidate Days

$\text{hw_day} = (\text{Temp Anomaly} \geq 2.0)$

Output:

- Boolean flag per row

✅ Step 2 — Identify Consecutive Sequences

A grouping variable is created:

$\text{hw_group} = \text{cumulative index where:}$
change in hw_day OR
change in station_id

This creates **continuous blocks of events**.

✅ Step 3 — Extract Event-Level Data

Group by:

$\text{station_id}, \text{hw_group}$

Compute:

Metric	Definition
start_date	first date
end_date	last date
duration_days	number of days
mean_anomaly	average anomaly
max_anomaly	peak anomaly

✔ Step 4 — Apply Duration Filter

KEEP only events where:
 $\text{duration_days} \geq 3$

4.4.4 Output Table: fact_Heatwaves

Column	Description
station_id	Station ID
hw_group	Event group
start_date	Start date
end_date	End date
duration_days	Duration
mean_anomaly	Avg intensity
max_anomaly	Peak intensity

✔ Observed Characteristics (from your analysis)

- Average duration $\approx 4\text{--}5$ days
- Max duration observed up to ~ 35 days
- Mean anomaly $\approx 4.7^\circ\text{C}$
- Extreme events exceed $+20^\circ\text{C}$ anomaly



4.5 Relationship Between Models

✔ Flow Dependency

Cleaned Temperature Data



Baseline Model



Anomaly Model



Heatwave Detection

✓ Data Relationships

Source	Target	Relationship
Baseline	Anomaly	1:Many (station-day)
Anomaly	Heatwaves	Derived (grouping)

⚠ 4.6 Model Assumptions

✓ Baseline Assumptions

- Climate stable enough over 30 years to define normal
- Each day-of-year has sufficient observations

✓ Anomaly Assumptions

- Mean Temp represents daily condition accurately
- Missing values have been correctly imputed

✓ Heatwave Assumptions

- Threshold = +2°C is valid across regions
- Minimum duration = 3 days captures extreme events

✓ 4.7 Limitations

- Local microclimates not modeled explicitly
- Baseline does not account for long-term shifts within period
- Heatwave threshold fixed (not adaptive by region)
- Wind not directly included in heatwave model

Excellent — this is the **final technical core section**, and it must clearly demonstrate:

- ✓ Advanced analytics capability
- ✓ Proper statistical modeling
- ✓ Forecast integration
- ✓ Alignment with your actual implementation

We will now build this as a **formal modeling specification**, not just explanation.

SECTION 5 — TREND ANALYSIS & FORECAST MODEL SPECIFICATION

5.1 Section Overview

This section defines the **statistical and predictive modeling components** of the Temperature and Wind Analytics module.

It includes:

- Long-term temperature trend analysis
- Monthly and station-level regression modeling
- Time-series forecasting using SARIMA
- Confidence interval estimation for predictions

These components transform anomaly datasets into **quantitative climate insights**, enabling:

- Detection of warming rates
- Comparison across stations and seasons
- Projection of future climate behavior



5.2 Temperature Trend Analysis

5.2.1 Purpose

The trend model quantifies the **rate of temperature change over time**, providing a measure of:

- Long-term warming
- Seasonal variability
- Station-level differences

5.2.2 Input Data

The model uses:

Temp Anomaly (°C)

instead of raw temperature, because:

- Removes seasonal effects
- Standardizes across stations
- Aligns with climate science best practices

5.2.3 Time Index Construction

To support regression modeling, a continuous time variable is created:

$$\text{time} = \text{Year} + (\text{Month} - 1) / 12$$

Example:

- January 2020 $\rightarrow$ 2020.00
- July 2020 $\rightarrow$ 2020.50

5.2.4 Regression Model

A **linear regression model** is applied:

$$A(t) = \beta_0 + \beta_1 \cdot t + \varepsilon$$

Where:

- $A(t)$ = anomaly at time t
- β_1 = slope (trend)
- ε = residual error

5.2.5 Trend Metric (Final Output)

To express the trend in a meaningful unit:

$$\text{Trend (}^\circ\text{C per decade)} = \beta_1 \times 10$$

5.2.6 Granularity of Trend Models

Trends are computed at:

✅ **Monthly level (per station)**

station_id + Month

✅ **Regional level (aggregated)**

5.2.7 Output Table: Dim_trendline_derived

Column	Description
station_id	Station
Month	Month index
Trend (°C / decade)	Warming rate
n_years	Data coverage

✅ Interpretation

Trend Value	Meaning
Positive	Warming trend
Negative	Cooling trend
Near zero	Stable climate

✅ Observed Results (from your system)

- Overall warming trend across Atlantic stations
- Stronger warming in specific months
- Variation across stations

🧠 5.3 Model Validation (Trend Analysis)

✅ Data Sufficiency Check

Minimum observations required per group:
 ≥ 2 data points

✅ Stability Check

- Outliers removed before regression
- Missing values handled

✅ Residual Behavior

- Residual variability indicates:
 - Natural climate fluctuations
 - Noise in observations

🧙 5.4 Forecasting Model (SARIMA)

5.4.1 Purpose

The forecasting model estimates **future temperature behavior (2026–2030)** based on historical patterns.

5.4.2 Model Type

The model used is:

SARIMA (Seasonal AutoRegressive Integrated Moving Average)

5.4.3 Model Structure

General form:

SARIMA(p, d, q)(P, D, Q, s)

Where:

Component	Meaning
p	autoregressive order
d	differencing order
q	moving average order
P, D, Q	seasonal components
s	seasonal period (e.g., 12 months)

5.4.4 Input Data

The model uses:

- Daily or aggregated temperature series
- Preprocessed, cleaned dataset
- Seasonal patterns preserved

5.4.5 Forecast Computation

Predictions are generated for each future time step:

$\hat{Y}(t)$ = model prediction

5.4.6 Confidence Interval

For uncertainty estimation:

$$\text{Lower Bound} = \hat{Y} - 1.96\sigma$$

$$\text{Upper Bound} = \hat{Y} + 1.96\sigma$$

Where:

- σ = standard deviation of residuals

5.4.7 Output Table: dim_sarima_forecast

Column	Description
Date	Forecast date
SARIMA_Prediction	Expected temperature
Lower_Bound	Lower estimate
Upper_Bound	Upper estimate

Interpretation

- Narrow band → high confidence
- Wide band → uncertainty increases over time

5.5 Forecast + Historical Integration

Combined Dataset

The system integrates:

Observed Data (2015–2025)

+

Forecast Data (2026–2030)

Use in BI

Supports visualization of:

- Actual vs predicted temperatures
- Trend continuation
- Seasonal behavior

5.6 Relationship Between Models

✅ Data Flow

Cleaned Temperature Data



Anomaly Model



Trend Model



Forecast Model

✅ Table Dependencies

Source	Target	Purpose
Anomaly	Trend	Regression input
Daily Temp	Forecast	Time-series input
Forecast	BI	Visualization

⚠️ 5.7 Model Assumptions

✅ Trend Model

- Linear relationship sufficient
- Noise is random (ϵ term)

✅ Forecast Model

- Historical patterns continue into future
- Seasonal cycles remain stable
- Residuals are normally distributed

⚠️ 5.8 Limitations

- Linear regression may miss nonlinear warming patterns
- SARIMA assumes stationary time series after differencing
- Forecast uncertainty increases with time horizon
- External factors (policy, environment) not explicitly modeled

✅ 5.9 Final Output of Analytical Layer

The system produces:

✅ Trend Data

- Station-level warming rates
- Monthly patterns

✅ Forecast Data

- Future temperature estimates
- Uncertainty bounds

✅ BI Integration

- Trend charts
- Forecast overlays
- Comparative station analysis

Perfect — this is your **final technical section**, and we will do it properly: This must clearly explain your **actual Power BI model**, including:

- ✅ Exact table roles
- ✅ Relationship types (not generic)
- ✅ Grain differences (daily vs monthly vs derived)
- ✅ Why your schema is advanced

SECTION 6 — BI DATA MODEL & RELATIONSHIP ARCHITECTURE

6.1 Section Overview

This section defines the **Business Intelligence (BI) data model architecture** used to support the Temperature and Wind Analytics module.

The model is implemented in **Power BI** and follows a **hybrid architecture**, combining:

- Star schema (core relationships)
- Snowflake elements (normalized dimensions)
- Multi-fact analytical structure

The design enables:

- Efficient querying
- Multi-granularity analysis
- Integration of derived analytical datasets

🔧 6.2 Model Architecture Type

✓ Hybrid Star–Snowflake Analytical Model

The schema is not a simple star model because:

- Multiple fact tables exist (not just one)
- Some dimensions are normalized (baseline table)
- Analytical tables (anomaly, heatwave, trends) act as **derived facts**

✓ Architectural Layers

1. Base Facts (Daily Data)
2. Aggregated Facts (Monthly Data)
3. Derived Analytical Facts (Anomaly, Heatwaves, Trends)
4. Supporting Dimensions (Station, Calendar, Baseline)
5. Forecast Layer (SARIMA Output)

6.3 Table Classification

✓ 6.3.1 Dimension Tables (Descriptive Data)

These tables provide context and slicing capability.

◆ Dim_Stations_All

Column	Description
Climate ID	Primary station identifier
Station Name	Station name
Province	Province
Latitude / Longitude	Geographic coordinates
Region	Derived grouping

✓ Used for:

- Spatial filtering
- Station-level comparisons

◆ Model Calendar

Column	Description
Date	Primary key
Year	Year

Column	Description
Month	Month
Season	Derived season

✓ Used for:

- Time filtering
- Trend analysis
- Forecast alignment

◆ Dim_Baseline_Temp

Column	Description
station_id	Station
Month	Month
Day	Day
Baseline Mean Temp	Reference value

✓ Purpose:

- Supports anomaly computation
- Acts as a **snowflake dimension**

✓ 6.3.2 Base Fact Tables (Daily Granularity)

◆ Fact_Daily_Temperature (Dim_Daily_temp_2015)

Grain:

1 station × 1 day

Column	Description
Climate ID	Station
Date	Date
Max Temp	Daily max
Min Temp	Daily min
Mean Temp	Computed
Heat Deg Days	Heating index

Column	Description
Cool Deg Days	Cooling index

◆ Fact_Daily_Wind

Grain:

1 station × 1 day (only when gust recorded)

Column	Description
Climate ID	Station
Date	Date
Dir of Max Gust	Wind direction
Spd of Max Gust	Wind speed

✓ 6.3.3 Aggregated Fact Tables (Monthly)

◆ Fact_Monthly_Temperature

Grain:

1 station × 1 month

Column	Description
Monthly Avg Temp	Mean of daily values
Monthly Max Temp	Max observed
Monthly Min Temp	Min observed

◆ Fact_Monthly_Wind

Column	Description
Monthly_Max_Gust	Max gust speed
Monthly_Mean_Direction	Mean direction

✓ 6.3.4 Derived Analytical Fact Tables

These are **not raw data tables** — they are computed outputs.

◆ Fact_Daily_Anomaly

Derived from:

Daily Temperature + Baseline

Column	Description
Date	Date
station_id	Station
Temp Anomaly	Computed value
Baseline Temp	Reference

◆ Fact_Heatwaves

Derived from:

Daily anomaly sequence grouping

Column	Description
start_date	Start
end_date	End
duration_days	Length
mean_anomaly	Avg intensity
max_anomaly	Peak

◆ Dim_trendline_derived

Column	Description
station_id	Station
Month	Month
Trend (°C/decade)	Trend value

◆ dim_sarima_forecast

Column	Description
Date	Future date
Prediction	Forecast temp
Lower_Bound	Lower CI

Column	Description
Upper_Bound	Upper CI

6.4 RELATIONSHIP ARCHITECTURE (CRITICAL)

6.4.1 One-to-Many Relationships (Primary)

Station Relationships:

Dim_Stations_All (1)
↓
All Fact Tables (Many)

Key:

Climate ID

Calendar Relationships:

Model Calendar (1)
↓
All Fact Tables (Many)

Key:

Date

6.4.2 Snowflake Relationships

Baseline Linking:

Dim_Baseline_Temp
↓
Fact_Daily_Anomaly

Key:

station_id + Month + Day

Why Snowflake?

- Baseline depends on **partial date (month/day)**
- Not directly linked by full date

✓ 6.4.3 Many-to-Many Relationships

Occur in:

- Trend tables across multiple years
- Heatwave table (aggregates many dates)

Example:

One station → multiple months

One month → multiple stations

✓ 6.4.4 Bridge (Derived) Relationships

Heatwave table is **not directly joined row-by-row**.

It is derived via:

station_id + grouped date sequences

This is effectively a:

✓ bridge relationship through temporal grouping

✓ 6.4.5 Multi-Granularity Relationships

The model supports multiple grains:

Table Type	Grain
Daily facts	Station × Day
Monthly facts	Station × Month
Trends	Station × Month
Heatwaves	Station × Event

✓ This enables:

- Drill-down analysis
- Aggregation flexibility

6.5 KEY DESIGN DECISIONS

Why NOT a single fact table?

Because:

- Different grains (daily vs monthly)
- Different semantics (measurement vs events)
- Different use cases (trend vs forecast)

Why surrogate keys exist?

Examples:

- dTemp_Key
- wind_key

Used for:

- Optimized joins
- BI performance
- Faster query execution

Why anomaly & heatwave are separate facts?

Because:

- They are **derived features**
- Not direct measurements
- Require independent analysis

6.6 MODEL USAGE IN BI

The model enables:

Time-based analysis

- Trends over years
- Seasonal patterns

✓ Spatial analysis

- Station comparison
- Regional clustering

✓ Event analysis

- Heatwave frequency
- Duration patterns

✓ Forecast analysis

- Expected temperature trends
- Uncertainty visualization

✓ 6.7 FINAL MODEL SUMMARY

The BI model is:

- ✓ Multi-layered
- ✓ Multi-granular
- ✓ Analytical (not just descriptive)
- ✓ Fully integrated with predictive modeling

Here is your **REFERENCES** section in proper APA style (7th edition), formatted professionally for academic submission.

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